Minnal-Q

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Real-life use cases for quantum emergency routing

The core question in every scenario: one vehicle, several people who need help, roads that change with traffic or disaster, so in what order do we go? Minnal-Q answers it with a QAOA circuit that weighs travel time, traffic and how critical each person is. The scenarios below show where this decision comes up in real life, especially in Kerala.

Readiness labels used in this document

Label	Meaning
DEMO TODAY	Fits the hackathon build: 3–4 stops, 9–16 qubits, runs on AerSimulator today
NEAR FUTURE	Same engine plus extra data (real road maps, blocked roads, time windows); still small per vehicle
FUTURE HARDWARE	Needs many more qubits or splitting into many small sub-problems; depends on larger quantum computers

We use these labels to stay honest: the hackathon demo proves the method on small cases, and the larger scenarios show where it leads.

Use-case scenarios

1. Mass-casualty road accident	DEMO TODAY
Situation	A bus overturns on NH-66 near Edappally. One ambulance arrives first; three or four injured people are spread along the road and nearby side lanes, with different injury levels.
The decision	Who to reach first and in what order, before taking everyone to the trauma centre.
Quantum role	QAOA orders the stops so the critical patient (severity 4) is reached first, using traffic-adjusted travel times. 3 victims → 9 qubits.
Inputs → Output	Victim locations + triage severity + time of day → visiting order and arrival time per patient
Who benefits	Critical trauma patients (the "golden hour"), ambulance crews, hospital emergency teams

2. Flood rescue	NEAR FUTURE
Situation	Heavy monsoon flooding, as in the 2018 Kerala floods. A rescue boat or high-clearance vehicle must pick up stranded people from several homes and bring them to a relief camp or hospital.
The decision	Pickup order when some routes are under water and people have different medical needs (elderly, pregnant, injured).
Quantum role	Same QUBO; flooded roads get a very high travel cost so QAOA avoids them. Severity weights give priority to the most vulnerable.
Inputs → Output	Rescue requests + flooded-road map + vulnerability level → pickup order and safe route
Who benefits	Stranded families, disaster response teams, district emergency operation centres

3. Landslide response	NEAR FUTURE
Situation	Landslides in a hilly district, like Wayanad in 2024, cut roads and isolate several sites. Teams must reach multiple locations as roads reopen.
The decision	Which sites a vehicle visits, in what order, as roads open and close during the day.
Quantum role	Re-run QAOA every time the road status changes (the "re-plan loop"). Blocked roads are high-cost edges; sites with more casualties carry higher weight.
Inputs → Output	Site list + casualty estimates + live road status → updated route for each vehicle
Who benefits	Survivors in isolated areas, NDRF and fire-and-rescue teams, hospitals receiving patients

4. Festival and pilgrimage crowds	DEMO TODAY
Situation	Large gatherings such as Thrissur Pooram or the Sabarimala season. Medical teams handle several emergencies at once (heat stroke, falls, cardiac events) inside heavily congested areas.
The decision	Order of patient visits when congestion changes by the hour.
Quantum role	Time-of-day traffic factors become crowd-density factors. QAOA picks the order that reaches the critical cases first through the least crowded paths.
Inputs → Output	Emergency calls + crowd-density map + severity → visiting order for each medical unit
Who benefits	Festival visitors and pilgrims, event medical teams, police control rooms

5. Blood and organ transport	NEAR FUTURE
Situation	A vehicle must collect rare blood units from two or three blood banks, or carry a donor organ, and reach the hospital within a strict time limit. Kerala has used police-cleared "green corridors" for organ transport.
The decision	Pickup order that keeps every item within its safe time window.
Quantum role	Time-critical items get high weight; time limits are added as penalty terms in the QUBO.
Inputs → Output	Pickup points + time limits + traffic → order and timing that keeps every item viable
Who benefits	Transplant and surgery patients, blood banks, hospitals

6. Home-care and palliative visits	NEAR FUTURE
Situation	Kerala's community palliative-care network sends nurses or a doctor to several elderly or bed-ridden patients' homes in a day; some calls are urgent.
The decision	Visiting order that handles urgent patients early while keeping total travel low.
Quantum role	Same engine: severity becomes urgency; the trade-off slider balances travel time against urgency.
Inputs → Output	Patient list + urgency + traffic → daily visiting route
Who benefits	Elderly and palliative patients, community nurses, primary health centres

7. Oxygen and medicine supply in a health crisis	NEAR FUTURE
Situation	During a surge like COVID-19 in 2021, a vehicle delivers oxygen cylinders or critical medicines to several hospitals and homes with different levels of need.
The decision	Delivery order when some patients cannot wait.
Quantum role	Each stop's need sets its weight; QAOA orders deliveries so the most urgent stops come first.
Inputs → Output	Delivery requests + need level + traffic → delivery order
Who benefits	Patients at home and in hospital, health departments, supply teams

8. Inter-hospital ICU transfers	DEMO TODAY
Situation	A critical-care ambulance moves several stable-but-serious patients between hospitals (for scans, surgery or free ICU beds) in one trip.
The decision	Order of hospital stops that meets the most urgent transfers first.
Quantum role	Hospitals become stops; transfer urgency becomes severity. Small enough for today's simulator.
Inputs → Output	Transfer requests + urgency + traffic → stop order
Who benefits	ICU patients, hospital networks, bed-management teams

9. City-wide ambulance fleet	FUTURE HARDWARE
Situation	A whole city's emergency ambulance service (for example Kerala's 108 network) handling dozens of calls and vehicles at once.
The decision	Which ambulance serves which calls, and in what order, for the whole fleet together.
Quantum role	Too large for today's devices in one piece. The path forward is to split calls between ambulances classically, then run one small QAOA per ambulance, which Minnal-Q already does. Bigger quantum computers could solve larger pieces together.
Inputs → Output	All live calls + all ambulances + traffic → fleet-wide routes
Who benefits	An entire city's population, emergency control rooms, health planners

Summary of all scenarios

#	Scenario	Stops per vehicle	Qubits	Readiness
1	Mass-casualty road accident	3–4	9–16	**Demo today**
2	Flood rescue	3–5	9–25	**Near future**
3	Landslide response	3–5	9–25	**Near future**
4	Festival and pilgrimage crowds	3–4	9–16	**Demo today**
5	Blood and organ transport	2–4	4–16	**Near future**
6	Home-care and palliative visits	3–5	9–25	**Near future**
7	Oxygen and medicine supply	3–5	9–25	**Near future**
8	Inter-hospital ICU transfers	2–4	4–16	**Demo today**
9	City-wide ambulance fleet	Dozens	Hundreds+	**Future hardware**

Qubits = (stops per vehicle)². Above about 25–30 qubits, exact simulation on a laptop becomes slow; those cases need quantum hardware or more splitting into smaller pieces.

How one quantum engine adapts to every scenario

The QAOA circuit and the Qiskit code stay the same. Only the numbers fed into the QUBO change:

What changes	In the QUBO	Example scenarios
Traffic or crowds	Travel-time multipliers on each road	City accidents, festivals
Blocked roads	Very high cost on that road, so it is never chosen	Floods, landslides
How critical a person is	Severity weight (1, 2, 4, 8)	All emergency scenarios
Time limits	Extra penalty if a stop is reached too late	Blood and organ transport
Changing conditions	Rebuild the QUBO and re-run QAOA (re-plan loop)	Landslides, live traffic
More vehicles	Split stops between vehicles classically, then one QAOA per vehicle	City-wide fleet

Beyond emergencies: the same idea elsewhere

Choosing the best order or allocation under pairwise costs is a common pattern, so the Minnal-Q approach (QUBO + QAOA + classical checks) carries over to other areas:

- **Logistics and last-mile delivery:** courier and grocery routes in busy cities.
- **EV charging:** deciding when and where vehicles charge (Challenge 4 at this event).
- **Energy:** choosing which power sources meet demand at the lowest cost (Challenge 3).
- **School and public buses:** stop order and timing for many routes.
- **Waste collection and street maintenance:** visit order for municipal vehicles.

Honest reality check

- Today, classical computers are faster for every scenario at the sizes we can run. We do **not** claim quantum advantage.
- Our demo proves the method works correctly on small, real-sized pieces (3–4 stops per vehicle), checked against the exact best answer.
- Real deployment would need real road data, live traffic feeds, integration with control rooms, and larger quantum hardware for the biggest cases.
- The value of Minnal-Q today is a **tested, quantum-ready pipeline** for decisions where every minute counts.

Pitch line

"Whether it's a bus crash on NH-66, a flood in Kuttanad or a crowd at Thrissur Pooram, the question is always the same: one vehicle, many people, which order? Minnal-Q answers it with a quantum circuit that puts the most critical person first. Today it runs on a simulator for one ambulance. Tomorrow, on bigger quantum computers, it can route a whole city's emergency fleet."